

Introduction to Statistical Investigations

Second Edition

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Chapter 5

Comparing Two Proportions

So begins Unit 2

- Previously, we examined single proportions/single means.
- Now, we'll look at 2 proportions.

Single Proportion Example

- In 2008, researchers observed car occupants (drivers and passengers), recorded whether they were wearing seatbelts.
- National Occupant Protection Use Survey (NOPUS) collects data nationwide by sending trained observers to randomly sampled roadways.
- They observe passenger vehicles between the hours of 7 a.m. and 6 p.m. while standing at the roadside or riding in a vehicle in traffic.
- Observers use hand-held clickers to count use of shoulder belts by drivers and right-front passengers.

	2008
Wearing seat belt	140,270
Not wearing seat belt	28,730
Total	169,000

Two Proportion Example

- Repeat the study in 2018. We get the following table:

	2008	2018	Total
Wearing seat belt	140,270	115,525	255,795
Not wearing seat belt	28,730	13,409	42,139
Total	169,000	128,934	297,934

What to do with this information?

- *Conditional proportions*
- Two options:
 - Column proportion: $140,270 / 169,000 = 0.83$ and $115,525 / 128,934 = 0.896$
 - Row proportion: $140,270 / 255,795 = 0.548$ and $115,525 / 255,795 = 0.451$

	2008	2018	Total
Wearing seat belt	140,270	115,525	255,795
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What can't you do with this information?

- Column proportion: $140,270 / 169,000 = 0.83$ and $115,525 / 128,934 = 0.896$
- Row proportion: $140,270 / 255,795 = 0.548$ and $115,525 / 255,795 = 0.451$
- Can't compare 0.83 and 0.896: Sample size is different.
- Also can't compare 0.548 and 0.451 for the same reason.

Moving Forward

- *Relative Risk*: Can help summarize the association: how many times greater risk of an outcome is for one group versus another.
- $RR = \frac{\widehat{p}_1}{\widehat{p}_2}$

Exploration: Murderous Nurse